

The Wisdom of Crowds: Applying Condorcet's Jury Theorem to Forecasting U.S. Presidential Elections[☆]

Andreas E. Murr^{*}

*Department of Politics and International Relations, University of Oxford, Manor Road Building,
Manor Road, Oxford, OX1 3UQ, UK.*

Abstract

Professional forecasters increasingly rely on citizen forecasts to predict election results. Following this approach, forecasters have been predicting the party to win which most citizens said will win. This approach predicts winners and vote shares well, but related research has shown that some citizens forecast better than others. Extensions of Condorcet's jury theorem suggest that naïve citizen forecasting can be improved by delegating the forecasting to the most competent citizens and by weighting their forecasts by their level of competence. Indeed, doing so increases the accuracy of vote share predictions and the number of correctly forecasted states. Allocating the state's electoral votes to the candidate who most weighted delegates say will win yields a simple, but successful forecasting model of the U.S. Presidency. The 'wisdom of crowds' model correctly predicts 8 out of 9 presidential elections. The results suggest that delegating and weighting provide an easy way to improve citizen forecasting.

Keywords: Citizen forecasting, Combining forecasts, Condorcet's jury theorem, Election forecasting, Election surveys, Weighting

[☆]Paper prepared for submission to the Special Issue, *Forecasting U.S. Presidential Elections: Rival Statistical Models*, of *The International Journal of Forecasting* edited by Michael Lewis-Beck and Drew Linzer.

^{*}Corresponding author.

Email address: andreas.murr@politics.ox.ac.uk (Andreas E. Murr)

1. Introduction

Democracies repeatedly ask citizens to elect the candidates, parties, or policies they want or think are best. Accordingly, considerable effort has been spent on forecasting the results of these elections. The efforts fall into two broad categories. Forecasting models in the first category focus on preferences. Forecasting models in this category usually elicit vote intentions in surveys before the election to predict vote choice on election day (e.g., Linzer, 2013). Or they apply a general theory of voting behaviour—such as the idea that voters reward government for a good economy and punish it for a bad one—to forecast how voters will behave in a particular election (e.g., Abramowitz, 2012).

Forecasting models in the second category focus on judgements. They elicit vote expectations before the election and predict the candidate to win which most citizens forecast will win (e.g., Lewis-Beck & Skalaban, 1989). Because these models follow the democratic process by letting the simple majority of citizens decide which candidate is predicted to win, their predictive accuracy informs the debate about democracy itself.

The initial studies on citizen forecasting tested whether citizen forecasts predicted at all, and they did. They predicted well in the U.S. when forecasting which candidate will win the Presidency (Lewis-Beck & Skalaban, 1989; Lewis-Beck & Tien, 1999) and which candidate will win a state (Rothschild & Wolfers, 2013; Graefe, 2014); they also predicted well in Great Britain when forecasting which party will win the Prime Ministership (Lewis-Beck & Stegmaier, 2011) and which party will win in each constituency (Murr, 2011).

More recent studies even show that citizen forecasts predicted better than other approaches at the state and national level. At the state level, Rothschild & Wolfers (2013) compared the relative accuracy of citizen forecasts and vote intentions. Based on data from the American National Election Studies (ANES) from 1952

to 2008, they found that citizen forecasts predicted more states correctly than vote intentions in 8 out of 10 presidential elections. Similarly, Miller et al. (2012) compared the relative accuracy of citizen forecasts and the prediction market *Intrade*. Using a unique survey during the 2008 U.S. presidential election, they showed that citizen forecasts predicted state outcomes more accurately than the prediction market.

At the national level, Graefe (2014) compared the relative accuracy of several approaches in predicting U.S. presidential elections. The comparison pitted citizen forecasting against quantitative models, vote intentions, prediction markets, and election experts. Comparing their relative accuracy in the seven elections between 1988 and 2012, he found that citizen forecasts performed best in predicting election winners and vote shares.

Although the above cited studies of citizen forecasting aggregated judgements, they implicitly followed two aggregation principles for preferences in democratic elections: universal suffrage and ‘one person, one vote’. The above cited studies followed these principles by using all citizen forecasts and by giving them equal weight. These two principles are indispensable when aggregating preferences; it can be beneficial to dispense with them, however, when aggregating judgements. Citizen forecasting can become even better by delegating the task of forecasting to the most competent citizens, and by weighting their forecast depending on their competence.

Both delegating and weighting require a measure of forecasting competence. The following study proposes a new method to measure it. The method identifies the good forecasters in the past, estimates their probability of being correct in the current election, and takes this estimate as a measure of their competence. It then delegates the forecasting task to the most competent members and weights their forecast by their competence. Delegating and weighting increases the aver-

age probability of a correct forecast. Accordingly, the method improves citizen forecasts without having to interview more citizens. Instead, it improves forecasts with available data. The reason for the success of delegating and weighting lies in Condorcet's jury theorem and its extension, which highlight when groups outperform individuals in binary choice tasks (Condorcet, 1785, 1994).

2. Linking Citizen Forecasts to Condorcet's Jury Theorem

Murr (2011) was the first to identify the link between citizen forecasting and Condorcet's jury theorem. The theorem features prominently in discussions about the advantages and disadvantages of democracy, but it applies to group decision making more generally. Condorcet derived the theorem for a group choosing between two alternative by simple majority vote, i.e. by choosing the alternative that receives the most votes of its members. He assumed that members vote independently of one another and that each member has the same chance of choosing the correct alternative.

The theorem implies that if each group member has a greater than .5 chance of voting for the correct alternative, then the probability of a correct simple majority vote approaches unity as the group size increases to infinity ('wisdom of crowds'). The reverse also holds. If each member has a smaller than .5 chance of voting for the correct alternative, then the probability of a correct simple majority vote approaches zero as the group size increases to infinity ('folly of crowds'). The above cited studies on citizen forecasting suggests, however, that citizens have a greater than .5 chance of correctly forecasting the election winner.

The aggregation effect emerges even in small groups. Consider a group of citizens who each correctly forecast 60 percent of the time. With 5 citizens the group correctly forecasts 68 percent of the time; with 25 citizens it does so 85 percent of the time. By contrast, with citizens who all correctly forecast 70 percent

of the time, the groups of 5 and 25 citizens correctly forecasts 84 and 98 percent of the time, respectively.¹

3. Generalising Condorcet's Jury Theorem

Condorcet considered groups facing a binary choice situation with members who have the same competence and vote independently of one another. Subsequent research generalised the theorem to situations with more than two alternatives, with members of heterogeneous competence, and with dependent votes. For the present context, the most important generalisations are those allowing for heterogeneous competence and dependent votes.

An important generalisation of Condorcet's jury theorem allows citizens to vary in their competence. Grofman (1978) demonstrated that Condorcet's jury theorem still holds if the competence of the group members is symmetrically distributed with a mean of greater than .5. This theoretical generalisation is important in practice because Lewis-Beck & Skalaban (1989) and Lewis-Beck & Tien (1999) have shown that indeed citizens vary in their forecasting competence.

It is also likely that the forecasts of citizens are correlated with each other. With random sampling it is unlikely that survey respondents have observed how the others forecasted or that they talked to each other to exchange information or to convince the other about who will win. Nevertheless respondents might share the same information if they were interviewed on the same day or if they have the same prior beliefs because of similar levels of political interest or because of their partisanship.

A potential cost of correlated votes is that they might decrease the effective

¹ The number of correct group members follows a Binomial distribution. With plurality voting, the probability of a correct group decision equals one minus the probability that less than half of the members make a correct decision. In R, the following code replicates the above calculation assuming odd group sizes: `n<-c(5,25,5,25)`; `p<-c(.6,.6,.7,.7)`; `1-pbinom((n+1)/2,n,p)`.

sample size. If every pairwise correlation is zero, then every citizen brings novel information to the group forecast; if, by contrast, every pairwise correlation is one, then all citizens have the same information, and interviewing one citizen is enough to know how the group would forecast.

Ladha (1992) specified the conditions under which Condorcet's jury theorem still holds when votes are correlated. He derived a formula to calculate how large the average correlation can be given the size of the group and the average competence so that the theorem still holds. To illustrate his result, consider a group of 10 members. With an average competence of .6 the average correlation can be up to .34; with an average competence of .9 the average correlation can be up to .82.² In other words, the average correlation can be quite high even for small groups of medium to high competence, and Condorcet's jury theorem still holds.

The formula also highlights when correlated votes can improve group forecasts. Ladha (1992, 629) gives an example in which adding incompetent citizens to a group of competent citizens increases the probability that the group will make the correct decision. In the example, the votes correlate positively within both groups, but negatively between both groups. The reason for the increase in group performance is that the addition decreases the average correlation. And the benefit of decreasing the average correlation outweighs the cost of a lower average competence. In fact, without the correlated votes the combined group would decide worse than with the correlated votes. Because citizen forecasting has been mostly accurate in the past, it follows from the assumptions that the average correlation is low and that the group-wise correlation patterns may even benefit citizen forecasting. Consequently, the following assumes that forecasts are uncorrelated to simplify the discussion and analysis.

² In R the following code replicates the numbers: `n<-c(10,10); p<-c(.6,.9); p-(n/(n-1))*(p-.25)*(1-p)/p.`

4. Improving Citizen Forecasts

Besides generalising Condorcet's jury theorem, several authors have used it as a framework for suggesting ways of improving group decision making. These suggestions roughly fall into three categories, focusing on increasing the group size, the competence of the group members, or the voting weight of competent members.

4.1. *Increasing the Group Size*

The first category of suggested improvements focuses on group size. It follows directly from Condorcet's jury theorem that if citizens forecast better than chance, then asking more citizens increases group performance. But the attractiveness of this suggestion varies with the level of aggregation that we are interested in.

Increasing the group size represents a better way of increasing state than national level forecasts. Surveys asking citizens to forecast which candidate will win the Presidency have large sample sizes regularly. Increasing the number of respondents will have little effect for them because Condorcet's jury theorem follows the law of diminishing returns. The rise in the chance of a correct group forecast by interviewing one more citizen becomes smaller with increasing sample size. Consider two groups whose members have an average probability of a correct forecast of .7 and .9, respectively. Both groups have the same chance of a correct forecast already with 50 members.

Surveys asking citizens to forecast which candidate will carry the state, by contrast, have small samples in each state. Interviewing more citizens in states with close elections will improve citizen forecasting. Because citizens correctly forecast the winning margin in each state (see below) citizen forecasts themselves can guide the decision where to collect more of them (Murr, 2011). If citizens forecast a close election in the state, surveys should sample more of them; if not,

surveys should stop interviewing citizens there and shift resources to states with close elections. In other words, the sampling design is not fixed prior to the survey, but adapts to responses collected during the survey (Thompson, 1992; Thompson & Seber, 1996).

Adaptive sampling will prove most useful to surveys who aim only at forecasting state election results. Surveys with multiple aims, such as national election studies, however, will not consider adaptive sampling. Some of their aims will conflict with the aim to maximise the predictive power of citizen forecasting. For instance, studying turnout requires equal numbers of respondents in states with narrow and wide winning margins, whereas forecasting elections could be improved by unequal numbers of respondents in those states, as outlined above. When adaptive sampling is not an option, the suggested improvements in the other two categories show ways to optimise citizen forecasting with a given group size.

4.2. Increasing the Average Group Competence

The second category of suggested improvements focuses on increasing the average group competence. One way to do so is to educate the citizenry, telling them what predictors work best and what decision rule produces the best forecast (Murr, 2013). Another, perhaps more immediately practicable, way is to delegate the task of making a forecast to the most competent members of the group. Kazmann (1973) proposed this idea, showing how the probability of a correct group decision changes by excluding members who fall below a moving threshold of competence. Accordingly, delegating the forecast to the most competent members moves away from ‘universal suffrage’ towards ‘conditional suffrage’.

In commenting on Kazmann’s proposal, Grofman (1975) presented a formula for calculating when an increase in the group average compensates for the decrease in group size. The formula shows that delegation improves group forecasts

if the benefit of an increased average competence outweighs the costs of a decreased group size. I label this approach ‘delegated citizen forecasting’.

4.3. Increasing the Voting Weight of Competent Members

The third category of suggested improvements focuses on aggregation rules. One way to increase group accuracy is to raise the level of agreement required for a group decision. Replacing the simple majority rule with a supermajority rule or even unanimity will make group decisions more accurate (Grofman & Owen, 1986). At the same time, however, the higher requirements will also increase the chance of gridlock, leading to more undecided choice situations.

A more promising way of increasing group accuracy dispenses with the democratic principle of ‘one person, one vote’. Instead of equal votes, the group members receive unequal votes. The weight of their vote increases with their level of competence. Shapley & Grofman (1984) and Nitzan & Paroush (1982) showed that the optimal decision rule assigns voting weights proportional to the logged odds of competence. Assuming that both choices are a priori equally likely to be correct and that votes are independent, the authors showed that given a group size and average competence, the proposed decision rule maximises the probability that the group will make the correct decision.

One reason why the weights maximise group performance is that more competent members have more votes. Another reason is that the weights tend to reverse incorrect decisions. The decision rule identifies incompetent members and then forecasts the opposite of what they said, thereby translating incompetent forecasters into competent ones. Because incompetent members choose the wrong alternative more often than not, reversing their choice picks the right alternative most of the time. I label this approach ‘weighted citizen forecasting’.

4.4. Determining the Competence of Members

Most of the suggestions to improve group forecasts assume knowledge of the member's competence. We cannot, however, directly observe it. Consequently, various ways to estimate competence have been proposed. Grofman & Feld (1986) discuss two approaches: asking citizens to assess the relative competence of each other (as proposed by Bodily, 1979, among others), or counting how often the citizen agrees with the majority opinion (Mirkin, 1979).

Because these studies were mainly concerned with preference aggregation, they lacked an external criterium against which to evaluate competence. Instead, they had to refer to auxiliary constructs such as the perceived competence or the agreement with the majority opinion. In contrast to that, the present study focuses on judgement aggregation and has the advantage of having an 'external truth'. When forecasting elections, we can compare forecasts to what actually happened.

The following study proposes a new method to measure forecasting competence. The method identifies the correlates of good and bad forecasters in past elections, and selects and weights forecasts in the current election accordingly. Optimally delegating the forecasting task to the competent citizens and optimally weighting their forecasts will increase the average probability of a correct individual forecast, and thereby increase the probability of a correct group forecast. The method is widely applicable because it relies only on information that most surveys collect anyway. I label this approach 'delegated and weighted citizen forecasting'.

5. Data

The citizen forecasts and the correlates of forecasting competence come from the Cumulative Time Series Data of the American National Election Studies (2010, 2012). The ANES draws national samples, but it does not draw samples to repre-

sent the states. This poses a problem for research trying to infer from the sample to the state population. The proposed forecasting model, however, does not aim to estimate a state's proportion of correct citizen forecasts. A representative sample would improve the model's accuracy only if the population in question forecasts better than the sample does. Indeed, this papers shows that delegating and weighting by past performance makes the sample even less representative, but at the same time makes citizen forecasting even better.

The Cumulative Data consist of pre-election surveys with usually half of the interviews already finished four weeks before election day. The ANES asked respondents the following question: "How about here in [R's state]. Which candidate for President do you think will carry this state?" Respondents that did not answer this question are excluded from the analysis.³ The ANES asked this question in eleven elections. The ANES asked it for the first time in 1952 and then in every election survey between 1972 and 2012, except for 2000. Comparing these citizen forecasts with historical election results reported by the American Presidency Project (2014) shows who was correct and who was not.

6. Forecasting the forecasters

The following proposes a new method to 'forecast the forecasters'. It finds the correlates of correct forecasts in the past to calculate the predicted probability of a correct forecast in the current election. It then takes this predicted probability as a measure of forecasting competence to optimally delegate and weight forecasts. Delegating and weighting in turn improve the probability of a correct group forecast.

Using the past to predict who will correctly forecast in the future poses a

³ The proportion of non-response decreased steadily from 23 percent in 1952 to 5 percent in 2012.

methodological challenge. The challenge is to select variables with predictive power (Ward et al., 2010a). In other words, the goal is not to select which variables were statistically significant in a past election. This result might not generalise to future elections. Instead, the goal is to select the variables that maximise the accuracy of out-of-sample predictions, such as the future.

One way to meet the challenge is cross-validation (Geisser, 1975; Efron, 1983). It randomly divides the data into two sets, a ‘training’ and a ‘test’ set. Using the training set it estimates several models, and tests their predictive power using the test set. Accordingly, division, estimation, and prediction represent the three steps of cross-validation. Repeating these three steps several times provides a measure of the models’ overall predictive power.

More specifically, I followed the procedure outlined by Ward et al. (2010a, 8) and performed a 4-fold cross-validation. In each election I randomly assigned respondents to one of four equal sized groups. Three groups form the training set to estimate the models, whereas the remaining group represents the test set for the out-of-sample predictions. The estimation starts with a full model, but subsequently drops variables if it increases predictive power. To measure the predictive power I used the area under the Receiver Operator Characteristic (ROC) curve.⁴ I repeated this procedure four times such that each of the groups serves in turn as the test set while the other three make up the training set. Because the results of this process depend on how the data was divided in the first place, the 4-fold cross-validation was repeated 10 times.⁵

⁴ The ROC curve represents the performance of a binary classifier as the discrimination threshold varies. It shows the trade-off between the rate of true positives (i.e., the number of accurately predicted correct forecasters over the total number of correct forecasters) and the rate of false positives (i.e., the number of inaccurately predicted correct forecasters over the total number of incorrect forecasters). The larger the area under the ROC curve, the larger the predictive power.

⁵ The code to implement the procedure builds on Ward et al. (2010b). They state that “[t]here is no evidence that the choice of [the number of folds] makes much difference in most applications, as long as the subsample in each fold is large enough to calculate the usual statistics.” (Ward et al.,

The success of the cross-validation exercise depends on which variables were included in the full model at the beginning of the procedure. To the best of my knowledge, there has been no study showing which variables correlate highly with the quality of forecasting at the state-level. So as a starting point I include the variables that Lewis-Beck & Tien (1999) consider as potentially relevant for the quality of forecasts at the national level. They include variables such as the perceived closeness of the election, whether a respondent forecasts the party to win with which she identifies or for which she intends to vote for, as well as measures of political involvement and socio-demographics. In addition to their list, I include the age of the respondents as well as a dummy variable scoring one for white respondents and zero otherwise. The Appendix provides more details on the variable coding. It turns out that what predicts correct forecasts differs between state and national forecasts, but a detailed discussion is beyond the scope of this paper.

Figure 1 shows which variables best predict a correct forecast in each election. Figure 1 shows that the same variables are included or excluded again and again. For instance, whether a respondent perceives the election to be close features in every model when it is available. The degree of media exposure, by contrast, never features in the selected model when it is available. In addition, the set of selected predictors overlaps greatly from one election to the next. For instance, the set of predictors in 1952 is identical to the set in 1972, except for one variable.

Table 1 shows the results of a binary logistic regression of whether or not the forecast was correct on the selected predictors for each year. Table 1 highlights that the direction of the relationship between the variables stays constant over time. For instance, those who perceive the election to be close forecast worse

2010a, footnote 5 on page 8). In the present case, the 2004 survey has the smallest fold size of 274 respondents with a total sample size of 1096 respondents.

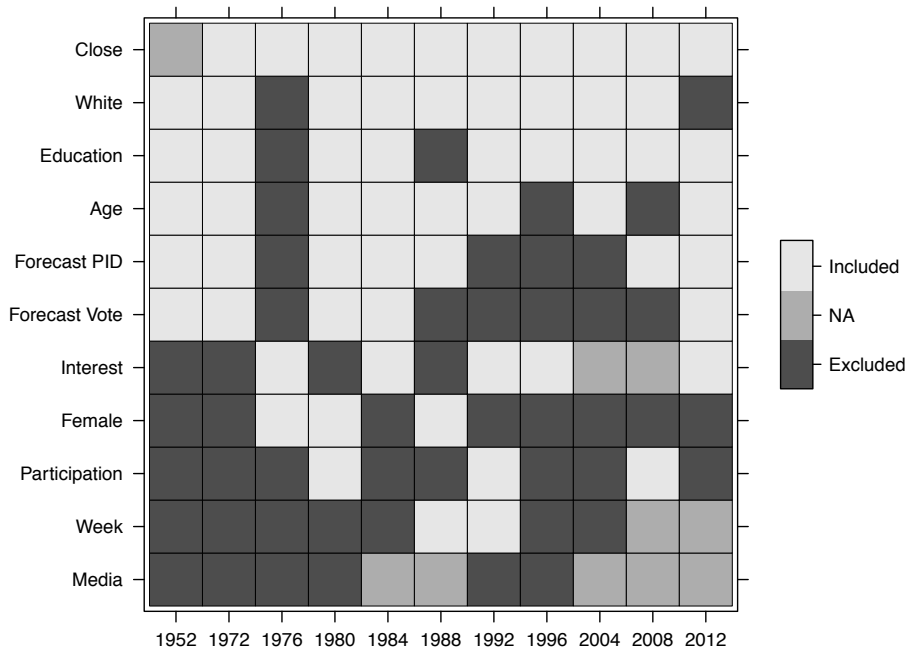


Figure 1: Selected predictors of a correct citizen forecast. The selection maximised the area under the ROC curve in a 4-fold cross validation exercise, which was repeated 10 times.

than others in every election. The higher educated always forecast better than the less educated. And so on. To sum up, the correlates of forecasting quality remain highly stable over time. Under these conditions delegating and weighting has the largest effect. The high stability in both the set of predictors and in their direction of correlation means that delegating and weighting will increase forecasting quality.

The next step of the forecasting model takes the estimates of Table 1 to calculate the forecasting competence of citizens. The calculation combines the respondents' values of variables in the current election with the corresponding regression estimates from the previous election. Using both it calculates the predicted probability of being correct, which serves as the estimated competence of the forecaster. The model then excludes the least competent citizens in a state if doing so increases the probability of a correct group forecast. Finally, the forecasting

Table 1: The effect of the selected predictors on making a correct forecast in each year, logistic regression results.

	Constant	Close	Educ.	White	Age	PID	Vote	Interest	Female	Partic.	Week	AUC
1952	−0.42		0.32	0.67	0.04	−0.77	0.30					0.65
<i>n</i> = 1436	(0.27)		(0.06)	(0.22)	(0.04)	(0.13)	(0.13)					
1972	0.62	−1.27	0.27	0.63	0.15	−1.70	1.20					0.77
<i>n</i> = 2212	(0.23)	(0.12)	(0.05)	(0.15)	(0.03)	(0.13)	(0.13)					
1976	0.26	−0.42						0.22	0.12			0.57
<i>n</i> = 1825	(0.18)	(0.10)						(0.07)	(0.10)			
1980	−0.22	−0.65	0.18	0.53	0.06	−0.81	0.20		−0.20	0.28		0.67
<i>n</i> = 1281	(0.26)	(0.12)	(0.06)	(0.17)	(0.03)	(0.13)	(0.13)		(0.12)	(0.08)		
1984	0.31	−0.60	0.25	0.85	0.07	−1.12	0.64	−0.19				0.69
<i>n</i> = 1891	(0.24)	(0.11)	(0.05)	(0.13)	(0.03)	(0.13)	(0.13)	(0.08)				
1988	1.11	−0.59		0.66	0.03	−0.36			−0.20		−0.09	0.63
<i>n</i> = 1720	(0.21)	(0.11)		(0.12)	(0.03)	(0.11)			(0.10)		(0.02)	
1992	0.13	−1.01	0.16	0.33	0.09			0.17		−0.01	−0.01	0.64
<i>n</i> = 2062	(0.24)	(0.11)	(0.04)	(0.11)	(0.03)			(0.08)		(0.07)	(0.02)	
1996	0.05	−0.66	0.14	0.34				0.18				0.61
<i>n</i> = 1608	(0.21)	(0.11)	(0.05)	(0.12)				(0.08)				
2004	0.57	−1.21	0.24	0.34	0.11							0.68
<i>n</i> = 1096	(0.28)	(0.16)	(0.07)	(0.16)	(0.04)							
2008	0.00	−0.55	0.26	0.23		0.03				0.21		0.63
<i>n</i> = 2073	(0.15)	(0.10)	(0.05)	(0.10)		(0.10)				(0.06)		
2012	0.93	−1.26	0.25		0.06	−0.24	−0.33	0.13				0.70
<i>n</i> = 5489	(0.15)	(0.07)	(0.03)		(0.02)	(0.08)	(0.08)	(0.05)				

Note: Standard errors in parentheses. *n* stands for sample size; AUC stands for area under the ROC curve. The variable coding is explained in the Appendix.

model uses the estimated competence to optimally weight the forecasts.

7. Forecasting states

Moving from naïve to delegated and weighted citizen forecasting increases the number of correctly predicted states. Table 2 compares the accuracy of naïve, delegated, weighted, and delegated and weighted citizen forecasting at both the individual and aggregate level. Table 2 shows that delegating and weighting improves citizen forecasts in every election. Without delegating and weighting, on average 69 percent of citizens get it right, whereas with both 76 percent do—an increase of 7 percentage points. This increase represents the effect of delegating and weighting at the individual level.⁶

The increase in the individual accuracy translates into greater aggregate accuracy. Table 2 shows that delegating and weighting improves state forecasts in 9 out of 10 elections; in the remaining election it makes the same predictions as naïve citizen forecasting. Overall, naïve citizen forecasting correctly predicts 82 percent of the states, whereas delegated and weighted citizen forecasting correctly predicts 87 percent of them—an increase of 5 percentage points. This increase represents the effect of delegating and weighting at the state level.

The comparison of individual and aggregated forecasts shows a ‘wisdom of crowds’ effect: groups outperform individuals. Although the performance varies across elections, groups usually outperform individuals. Consider delegated and weighted citizen forecasting. Overall, 76 percent of citizens get it right, whereas 87 percent of groups do—an increase of 11 percentage points. This increase embodies the ‘wisdom of crowds’ effect.

⁶ The proportion of delegates varies across years in the following way: 1972 (73 percent), 1976 (56 percent), 1980 (81 percent) 1984 (42 percent), 1988 (62 percent), 1992 (71 percent), 1996 (63 percent), 2004 (77 percent), 2008 (77 percent), and 2012 (77 percent).

Table 2: The percentage of correct individual and aggregated forecasts of which presidential candidate will carry the state. In both cases, delegated and weighted forecasts outperform naïve ones.

Year	Individual level				State level			
	Delegated & Weighted				Delegated & Weighted			
	Naïve	Weighted	Delegated	Weighted	Naïve	Weighted	Delegated	Weighted
1952	0.64	—	—	—	0.71	—	—	—
1972	0.77	0.83	0.84	0.86	0.95	0.97	1.00	1.00
1976	0.63	0.63	0.65	0.65	0.82	0.76	0.82	0.84
1980	0.54	0.58	0.55	0.58	0.49	0.56	0.49	0.59
1984	0.70	0.74	0.80	0.83	0.87	0.97	0.90	0.93
1988	0.64	0.70	0.71	0.71	0.87	0.87	0.90	0.87
1992	0.70	0.74	0.73	0.75	0.91	0.97	0.97	0.97
1996	0.66	0.71	0.70	0.73	0.76	0.81	0.79	0.81
2004	0.76	0.82	0.80	0.83	0.90	0.90	0.93	0.93
2008	0.66	0.71	0.67	0.72	0.82	0.82	0.85	0.85
2012	0.75	0.81	0.78	0.82	0.96	0.94	0.96	0.96
Overall	0.69	0.74	0.74	0.76	0.82	0.85	0.86	0.87

Note: At the state level, I count the number of citizens who say that a candidate will win. The naïve version simply predicts the candidate with the highest number to win. The weighted version weights each citizen by past performance and then predicts the candidate with the highest weighted number to win. The delegated version drops the least competent citizens from the sample if doing so increases the chance of a correct group forecast. It then predicts the candidate with the highest number of citizens to win. Finally, the delegated and weighted version predicts the candidate to win who has the highest weighted number among the delegates.

Figures 2 and 3 map the coverage and performance of the ‘wisdom of crowds’ model for each election. Because the ANES used to interview citizen from more populous states, the model does not make predictions for some states most of the time. For instance, the ANES interviewed citizens of Montana for the first time in 2012.

8. Forecasting the U.S. Presidency

Citizen forecasting with delegation and weighting gets most of the states right, but it still could get the Presidency wrong. The ANES usually left out some states, and the model may correctly forecast states with few electoral votes, but incorrectly do so in states with many electoral votes. Partial coverage and wrong forecasts in populous states could lead the ‘wisdom of crowds’ model to forecast the wrong candidate to win the Presidency. To the best of my knowledge, no study has shown yet that citizens can forecast the Presidency based on electoral votes. It turns out that they can. The model performs well across changing political contexts. It correctly predicts the winner in elections with and without the incumbent running again, in elections of re-alignment or de-alignment, and in elections with traditional or modern campaigns.

Figure 4 shows that the ‘wisdom of crowds’ model correctly predicts 8 out of 9 Presidencies for which data was sufficiently available to call a winner. Figure 4 plots the forecasted number of electoral votes for each candidate in each election. It also plots the number of electoral votes which the model could not allocate, either because the race was tied or no respondents were interviewed in the state. Figure 4 shows only one tied race in 1952. As it is the first election, no weighting was available to predict the race.

The ‘wisdom of crowds’ model allocates the state’s electoral votes to the candidate who receives the highest number of weighted forecasts from the delegates;

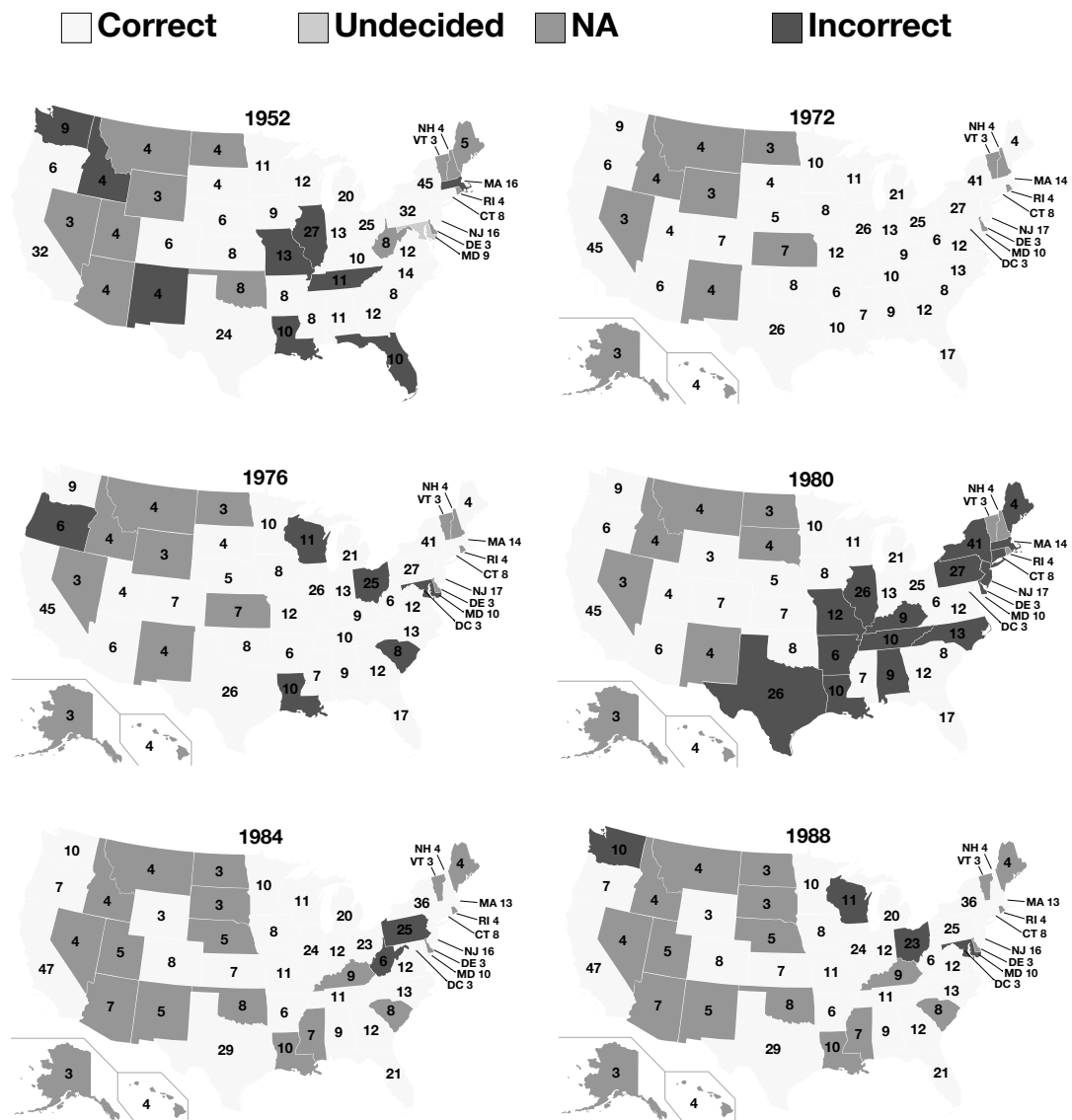


Figure 2: Map of forecasting performance by state from 1952 to 1988. The numbers indicate the state's electoral votes.

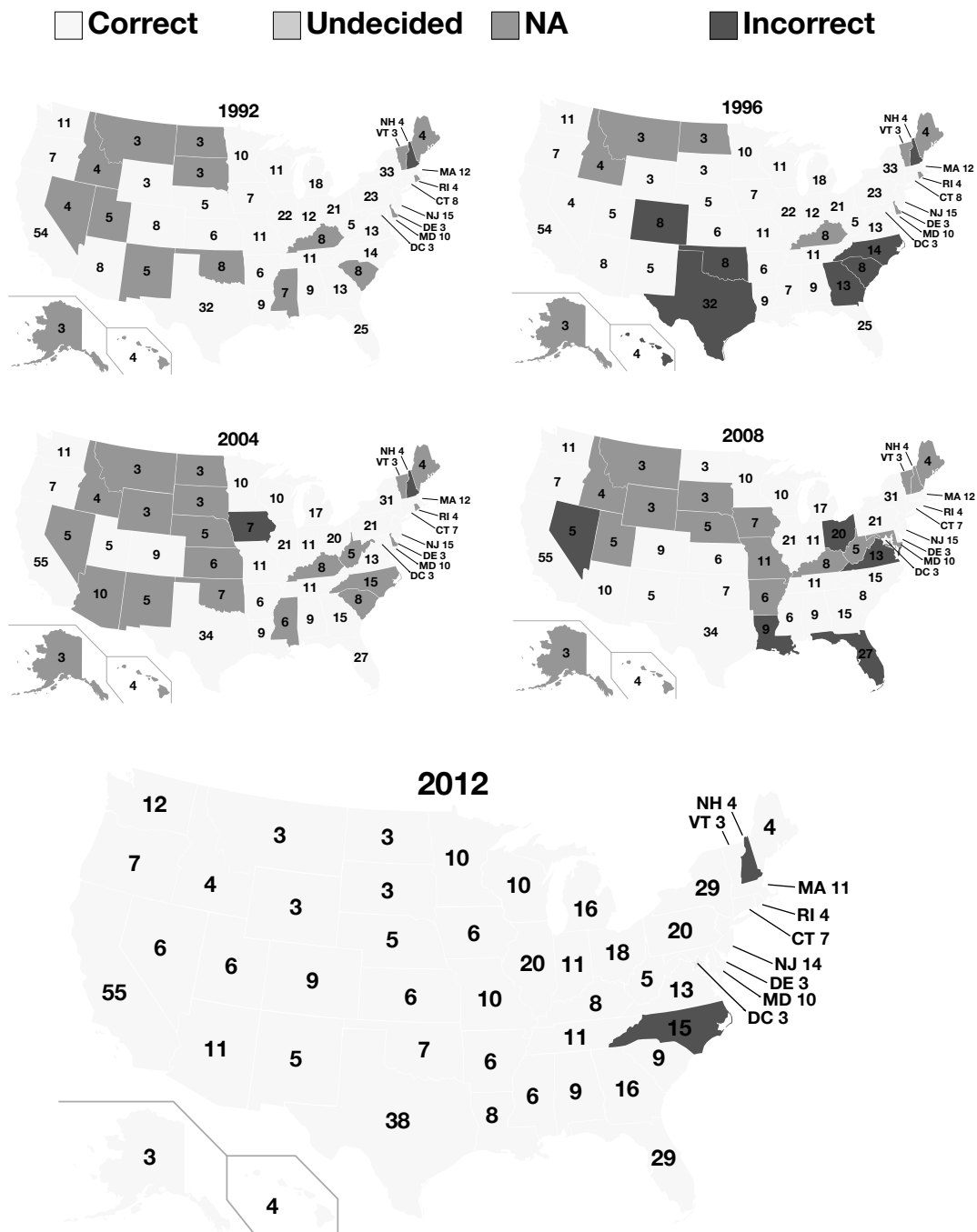


Figure 3: Map of forecasting performance by state from 1992 to 2012. The numbers indicate the state's electoral votes.

it then predicts the candidate who receives more than half of the total number of electoral votes to win the Presidency. The forecasting rule predicts exceptionally well. It correctly predicts 8 out of 9 elections.⁷ The only exception is in 1980, when it incorrectly predicts a Carter re-election. In 1976 and 2004 the model lacks sufficient data to make a forecast; no candidate crosses the threshold of at least 270 electoral votes. Because the ANES will continue to sample from all 50 states in the future, as it did in 2012, I expect that the same problem will not occur in future. To sum up, when a forecast is possible, the ‘wisdom of crowds’ model correctly forecasts 8 out of 9 elections.

9. Forecasting vote shares

The ‘wisdom of crowds’ model also accurately forecasts candidates’ votes shares in states. By relying on delegated and weighted citizen forecasts instead of the naïve ones, the model’s accuracy increases again.

For translating citizen forecasts into vote share forecasts at the state level the model needs historical data. A simple way to estimate the relationship between the two is by a linear regression (Lewis-Beck & Tien, 1999). The model regresses a party’s actual share of the two-party vote on the (weighted) forecast score, the balance of people who said that the party would carry the state. The measure of balance ranges between –1 ‘everybody said that the Republican party would win’ and 1 ‘everybody said that the Democratic party would win’.

The regression of vote shares on forecast scores represents a vote equation. Table 3 shows the vote equations for the Democratic Party. The vote equations are estimated for each year based on all past elections and the current one. Table 3 lists the cumulative vote equations separately for naïve as well as delegated and

⁷ Allocating the state’s electoral votes to the candidate who most citizens say will win yields the same results.

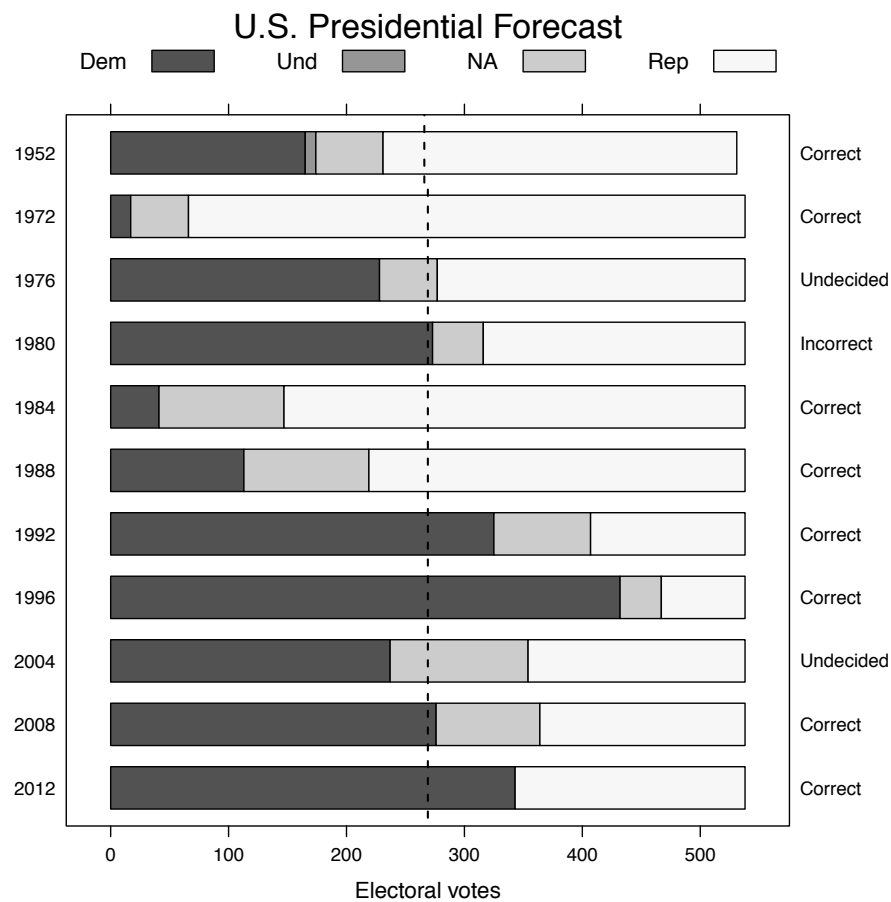


Figure 4: When data is sufficiently available to make a prediction, the ‘wisdom of crowds’ model correctly predicts 8 of 9 elections. The dashed line indicates how many electoral college votes a party needs to win the Presidency. The model forecasts the party which crosses the line to win the Presidency. Results are based on optimally delegated and weighted citizen forecasts, except in 1952 when the naïve ones are used.

weighted forecast scores. The results for the Republican Party are identical (no table shown).

The results show that moving from naïve to delegated and weighted citizen forecasting explains more of the variance in the vote shares. The last row in Table 3 shows that when pooling all elections, naïve citizen forecasting explains 68 percent of the variance, whereas the delegated and weighted version explains 72 percent—an increase by 4 percentage points.

The vote equations summarise how forecast scores correspond to vote shares. The last row in Table 3 shows that increasing the balance of weighted delegates who say that a candidate will win by 10 percentage points corresponds to an increase in his vote share of 1.3 percentage points. If the weighted delegates split equally between Republicans and Democrats, the Democratic candidate will receive 49 percent of the two-party vote.

These vote equations are based on in-sample estimation. The next step is to compare the out-of-sample predictive accuracy of delegated and weighted citizen forecasting with the naïve one. To do so, I calculate the vote share forecast for each election relying only on data that was available prior to the election. In other words, I combine the citizen forecasts for the current election with the vote equation of the previous one. For instance, to generate vote share forecasts for the 1984 election, I use the forecast scores from that year and combine them with the vote equation of 1980.

Table 4 highlights that in out-of-sample predictions of vote shares, delegated and weighted citizen forecasting outperforms the naïve version. Table 4 presents four measures of forecasting performance: the root mean squared error, the mean absolute error, which version was closer to the actual vote shares, and whether the candidate with the highest predicted vote share actually had the highest vote share. All four measures show that delegating and weighting increases forecasting

Table 3: The relationship between a candidate's share of the two-party vote and the balance of people who said that he will carry the state (forecast score). In every election the delegated and weighted forecast scores fit the data better than the naïve ones.

	Naïve			Delegated & Weighted		
	Intercept	Forecast score	R^2	Intercept	Forecast score	R^2
1972	0.45	0.14	0.24	0.50	0.18	0.40
$n = 38$	(0.02)	(0.04)		(0.03)	(0.04)	
1976	0.48	0.18	0.66	0.50	0.16	0.70
$n = 76$	(0.01)	(0.01)		(0.01)	(0.01)	
1980	0.47	0.15	0.61	0.49	0.14	0.65
$n = 115$	(0.01)	(0.01)		(0.01)	(0.01)	
1984	0.46	0.14	0.58	0.49	0.14	0.64
$n = 145$	(0.01)	(0.01)		(0.01)	(0.01)	
1988	0.47	0.14	0.57	0.49	0.13	0.62
$n = 175$	(0.00)	(0.01)		(0.00)	(0.01)	
1992	0.47	0.15	0.64	0.49	0.13	0.69
$n = 208$	(0.00)	(0.01)		(0.00)	(0.01)	
1996	0.48	0.15	0.67	0.49	0.13	0.72
$n = 250$	(0.00)	(0.01)		(0.00)	(0.01)	
2004	0.48	0.15	0.68	0.49	0.13	0.72
$n = 279$	(0.00)	(0.01)		(0.00)	(0.00)	
2008	0.48	0.15	0.68	0.49	0.13	0.72
$n = 313$	(0.00)	(0.01)		(0.00)	(0.00)	

Note: The unit of analysis is states. Estimates come from an OLS regression based on all previous elections and the current one. In each regression, states are weighted the number of respondents (Naïve) or by the sum of the absolute weight of each delegate's forecast (Delegated & Weighted). Standard errors in parentheses. The reported estimates are for Democratic candidates, but for Republican candidates they are the same.

accuracy. Overall, it decreases the root mean squared error (from 6.6 to 6.4 percentage points), the mean absolute error (from 4.5 to 4.3 percentage points), the distance between predicted and actual vote (in 53.4 percent of the cases), and the number of times the losing candidate has a higher predicted vote share than the winning candidate (from 17.2 to 15.3 percent of the cases). Moving from naïve to delegated and weighted citizen forecasts improves, therefore, predictions about who wins and by how much.

10. Why can groups forecast?

The ‘wisdom of crowds’ model correctly forecasts most of the time. More often than not, it correctly forecasts which candidate will carry a state, and which candidate will win the Presidency. But sometimes it gets it wrong. When it was able to forecast, it picked the wrong candidate in 56 out of 399 states, and in 1 out of 9 national elections. What explains these wrong forecasts?

Condorcet’s jury theorem explains when citizen forecasting predicts well. The explanation highlights the joint effect of the size of the group and the ability of its members. For instance, a high average probability of a correct forecast together with a large group size will correctly forecast most elections.

The group average derives from task and group characteristics (Murr, 2011, 777–781). The task that groups face is to correctly forecast which candidate will win. As the group’s estimated competence increases, so does its probability of correctly forecasting which candidate carries the state. Yet the difficulty of this task varies greatly. A switch from one party to the other is more difficult to forecast than stability. Similarly, a viable third-party candidate makes the task more difficult. Further, a large absolute change in the two-party vote at the state or at the national level might affect the difficulty of the task. Finally, a close election is difficult to forecast too. Few people should be able to do it, leading to a low group

Table 4: Out-of-sample predictions of vote shares. Delegated and weighted citizen forecasting predicts more accurately than the naïve one.

	Root mean squared error		Mean absolute error		Closest forecast		Winner correct	
	Naïve	Delegated & Weighted	Naïve	Delegated & Weighted	Naïve	Delegated & Weighted	Naïve	Delegated & Weighted
1976	7.9%	7.2%	6.2%	5.0%	34.2%	65.8%	68.4%	84.2%
1980	5.9%	6.8%	4.9%	5.8%	66.7%	33.3%	61.5%	56.4%
1984	4.3%	4.5%	3.0%	3.6%	63.3%	36.7%	93.3%	93.3%
1988	4.4%	4.1%	3.3%	3.4%	43.3%	56.7%	86.7%	90.0%
1992	4.3%	3.4%	3.5%	2.6%	33.3%	66.7%	87.9%	97.0%
1996	8.8%	8.3%	5.7%	5.4%	38.1%	61.9%	83.3%	83.3%
2004	6.9%	6.8%	4.5%	4.4%	55.2%	44.8%	89.7%	93.1%
2008	7.7%	7.4%	5.0%	4.7%	47.1%	52.9%	85.3%	79.4%
2012	6.2%	6.1%	3.9%	3.7%	43.1%	56.9%	92.2%	90.2%
Overall	6.6%	6.4%	4.5%	4.3%	46.6%	53.4%	82.8%	84.7%

Note: The unit of analysis is state. In each prediction, the state is weighted by the number of respondents (Naïve) or by the sum of the absolute weight of each delegate's forecast (Delegated & Weighted). The reported measures of fit are for Democratic candidates, but for Republican candidates they are the same.

average.

Besides its average competence and size, another important group characteristic is its proximity to election day and its informational diversity. Forecasting elections with lead time is difficult. As election day comes closer, people have a better idea about who will win. Similarly, when citizens forecast at different days, they add different types of information. Independent information helps citizen forecasting. In other words, states with more dispersed interview dates forecast better.⁸

Table 5 reports the results of a multilevel logistic regression of a correct group forecast (0 = no; 1 = yes) on the nine predictors. The model allows the intercept to vary by states and election years. Table 5 lists the results of three models. Model 1 includes all variables, but excludes the 2012 survey because its interview dates were unavailable. Model 2 keeps the original sample, but excludes interview dates from the model specification. Model 3 then includes the 2012 survey. Comparing the models shows that the size of the coefficients remain stable after excluding interview dates and including the 2012 survey.

Table 5 shows that the size and competence are important group characteristics and that the winning margin and swing state status are important task characteristics. All four variables affect the chance of a correct group forecast in expected ways. Groups struggle to forecast states switching from one party to the other, and to forecast close elections. But larger group and more competent groups increases the chance of a correct group forecast in any election. In other words, favourable group characteristics can compensate for unfavourable task characteristics.

In contrast to that, the absolute change in the two-party vote at both the state

⁸ I thank an anonymous reviewer for suggesting to add the following variables to the original model specification: state switching from one party to the other, the absolute change in the two-party vote at the state and at the national level, and the share of third party candidates.

Table 5: Predictors of a correct state forecast. Results are based on multilevel binary logistic regressions, with varying intercepts for states and election years.

	Model 1	Model 2	Model 3
(Intercept)	1.46 (1.53)	1.12 (1.28)	1.16 (1.22)
Weighted group size (log scale)	0.56 (0.30)	0.47 (0.24)	0.50* (0.22)
Median weight	1.36 (0.78)	1.35 (0.75)	1.47* (0.75)
State switches (0=no; 1=yes)	-2.24* (0.51)	-2.23* (0.51)	-2.32* (0.50)
Winning margin (log scale)	0.56* (0.19)	0.55* (0.19)	0.59* (0.18)
Abs. change in two-party vote (state)	7.28 (5.35)	7.18 (5.35)	7.68 (5.42)
Abs. change in two-party vote (nation)	-2.52 (12.28)	-1.33 (11.80)	-3.35 (10.98)
Vote share third party candidate	1.75 (5.39)	1.64 (5.24)	2.07 (5.09)
Days until election (median)	-0.00 (0.03)	— —	— —
Days until election (std. dev.)	-0.03 (0.06)	— —	— —
σ^2_{state}	0.00	0.00	0.00
σ^2_{year}	0.76	0.73	0.64
Log-likelihood	-88.78	-88.91	-94.18
Deviance	177.56	177.82	188.35
AIC	201.56	197.82	208.35
BIC	246.52	235.28	247.33
N	313	313	364

Note: The unit of analysis is states. The 2012 ANES data set does not include the response date yet. Standard errors are in parentheses. *indicates significance at $p < 0.05$. Group size and winning margin are logged to normalise their distribution.

and the national level, the vote share of third party candidates, and the interview date statistics are unimportant. The interview dates are measured as the number of days until the election. Both the state's median and variance of the interview dates fail to influence the chance of a correct group forecast. A state with a low median has the same chance as a state with a high one; a state with a dispersed interview dates has the same chance as a state with assembled interview dates. In short, only state switching, winning margin, group size, and group competence matter for citizen forecasting.

To calculate the magnitude of their effect on the probability of a correct forecast and to illustrate the uncertainty in the regression coefficients, I turn to statistical simulation (Imai et al., 2009). Using Model 3 in Table 5, the simulation takes 10,000 draws from the assumed normal distribution of the coefficients and combines them with the values of hypothetical groups to calculate their predicted probabilities and first differences.

I first calculated the predicted probability of a correct group forecast in a hypothetical election in Oregon in 1992. I specified Oregon as a swing state for presentational purposes, and set all independent variables to their median values in such swing states. In this hypothetical election, we would predict that the group has a 69 percent chance of making a correct forecast. I then calculated how the predicted probability would change as the independent variables change one after the other from their median to their maximum value. Table 6 shows the results.

All four variables have a noticeable effect on the probability of a correct group forecast. Increasing the weighted group size from 30 to 163 members raises the chance of a correct forecast by 14 percentage points. The median competence also has an effect. As the median competence moves from .74 to .91, the chance of a correct group forecast increases by 23 percentage points. The largest effect has the swing state status. In a stable state the chance of a correct group forecast is 26

Table 6: Changes in the expected probability of a correct group forecast as values of the independent variables change (Model 3).

Independent variable (change)	Expected change in pr. of correct forecast
Weighted group size (30 → 163)	0.14*(0.06)
Median competence (.74 → .91)	0.23*(0.10)
State switches (1 → 0)	0.26*(0.08)
Winning margin (.06 → .59)	0.19*(0.06)

Note: The varying intercepts are set to their median values, which corresponds to Oregon in 1992. The parentheses next to the name of each independent variable show the change in the value of each independent variable that corresponds to the change in the predicted probability of a correct group forecast from the baseline case. The baseline case for these calculations is the number to the left of each arrow in the parentheses. It corresponds to a predicted probability of a correct group forecast of 69 percent. Standard deviations for each expected change in the predicted probability are presented in parentheses next to each number. * $p < 0.05$.

percentage points higher than in a swing state. Finally, the winning margin also makes it more likely that groups will correctly forecast. Changing the winning margin from its median to its maximum value increases the likelihood of a correct forecast by 19 percentage points.

11. Conclusion

This paper has demonstrated the predictive power of a novel model to forecast who wins the U.S. Presidency, building upon the literature on citizen forecasting and on Condorcet's jury theorem. Graefe (2014) as well as Rothschild & Wolfers (2013) already have shown that most citizens correctly forecast which candidate will carry the state. They have not shown, however, that these forecasts also predict which candidate becomes President. This paper has taken the next step by allocating the state's electoral votes to the candidate who most citizens say will

win, and by predicting the candidate who receives more than half of the electoral votes to win the Presidency. When data was sufficiently available to make a forecast, the model correctly predicted 8 out of 9 elections between 1952 and 2012. It reliably forecasted the winner in changing political contexts, with just about 30 respondents on average interviewed in each state.

This paper also has shown for the first time that delegating the forecasting task to the most competent members and weighting their forecasts by past performance increases predictive accuracy. The paper identified good forecasters in the past to estimate the competence of forecasters in the current election. It then optimally delegated the forecasting and optimally weighted the forecasts. Together, delegating and weighting increased the share of correct forecasters in a state, thereby increasing the share of correctly forecasted states. Besides improving forecasts about who will carry the state, delegated and weighted citizen forecasting also improved predictions of candidates' votes shares. On a series of measures, delegated and weighted citizen forecasting outperformed the naïve version. In sum, delegating and weighting provide an easy way to increase the predictive power of citizen forecasting.

The next step is to adapt the general idea of delegating and weighting citizen forecasts to the national level and to other countries. The paper focused on citizen forecasts about which candidate will carry the state in U.S. presidential elections. In this case, the set of variables identifying good and bad forecasters stayed more or less constant from one election to the next, and the direction of the relationship between the variables and the quality of the forecast stayed constant most of the time too. These specifics will vary in other contexts. For instance, in another context the same variables may correlate with the quality of the forecast, but the direction of their correlation may change over time—in one election they correlate with a correct forecast; in the next they correlate with an incorrect one. The

approach presented here allows us to use these patterns to increase the accuracy of citizen forecasting in other contexts as well.

12. Appendix

Variables are coded as follows:

- *Age* (Age of respondent in years).
- *Close* (0 = respondent believes that one candidate will win by quite a bit; 1 = respondent believes that the race will be close).
- *Correct* (0 = respondent incorrectly forecasted the winning candidate; 1 = respondent correctly forecasted the winning candidate)
- *Education* (1 = Less than high school credential; 2 = High school credential; 3 = Some post-high-school, no bachelor's degree; 4 = Bachelor's degree; 5 = Graduate degree).
- *Female* (0 = respondent is male; 1 = respondent is female).
- *Forecast PID* (0 = respondent does not identify with a party or respondent does not identify with the party of the forecasted winner; 1 = respondent identifies with the party of the forecasted winner).
- *Forecast Vote* (0 = respondent has no vote intention or respondent forecasts a different candidate to win than the one for whom she intends to vote; 1 = respondent forecasted the candidate to win for whom she also intends to vote).
- *Interest* (1 = respondent is not much interested in political campaigns; 2 = respondent is somewhat interested in political campaigns; 3 = respondent is very much interested in political campaigns).
- *Media* (an additive index ranging from 1 to 5, in which respondents were assigned a score depending on whether or not they reported having watched

news about the campaign on television, heard something about the race on the radio, read about the campaign in magazines, or in newspapers).

- *Participation* (an additive index ranging from 0 to 4, in which respondents were assigned a 1 for doing any of the following acts of participation: contributing money, working for a candidate, trying to influence others, or attending a political meeting).
- *Week* (date of interview, 1 = one week before the election; 2 = two weeks before the election; 3 = three weeks before the election; 4 = four weeks before the election; 5 = five weeks before the election; 6 = six weeks before the election; 7 = seven weeks before the election; 8 = eight weeks before the election; 9 = nine weeks before the election).
- *White* (0 = respondent describes herself as black or ‘other’; 1 = respondent describes herself as white).

Acknowledgement

Previous versions of this paper have been presented at the 2014 MPSA conference, the 2014 Election Studies conference at the Center for Advanced Studies at the LMU München, and in seminars at the Universities of Essex, Konstanz, Nottingham, and Montevideo. I thank the participants for helpful comments and questions. I also thank Andreas Graefe, Simon Moritz, Ericka G. Rascón Ramírez, Peter Selb, and Paul Whiteley for useful discussions. Special thanks to Michael Lewis-Beck and Drew Linzer for inviting this manuscript, and in particular to Michael Lewis-Beck for his comments and encouragement while I was writing it. I am very grateful to the anonymous reviewers for their comments, which greatly improved the argument and presentation of the paper. All remaining errors are my own. The code to replicate the findings is available upon request.

- Abramowitz, A. (2012). Forecasting in a polarized era: The time for change model and the 2012 presidential election. *PS: Political Science & Politics*, 45, 618–619.
- American National Election Studies (2010). Time Series Cumulative Data Files. URL: <http://www.electionstudies.org/studypages/cdf/cdf.htm>.
- American National Election Studies (2012). The ANES 2012 Time Series Study. URL: http://www.electionstudies.org/studypages/anes_timeseries_2012/anes_timeseries_2012.htm.
- American Presidency Project (2014). Presidential elections data. URL: <http://www.presidency.ucsb.edu/elections.php>.
- Bodily, S. E. (1979). A delegation process for combining individual utility functions. *Management Science*, 25, 1035–1041.
- Condorcet, M.-J.-A.-N. d. C., Marquis de (1785). *Essai sur l'application de l'analyse a la probabilité des descisions redues a la pluralité de voix*. Paris: De l'Imprimerie Royale.
- Condorcet, M.-J.-A.-N. d. C., Marquis de (1994). Essay on the application of probability analyses to decisions returned by a plurality of people. In I. McLean, & F. Hewitt (Eds.), *Condorcet: Foundations of Social Choice and Political Theory* (pp. 11–36). Aldershot: E. Elgar. (Original work published 1785).
- Efron, B. (1983). Estimating the error rate of a prediction rule: Improvement on cross-validation. *Journal of the American Statistical Association*, 78, 316–331.
- Geisser, S. (1975). The predictive sample reuse method with applications. *Journal of the American Statistical Association*, 70, 320–328.
- Graefe, A. (2014). Accuracy of vote expectation surveys in forecasting elections. *Public Opinion Quarterly*, 78, 204–232.
- Grofman, B. (1975). A comment on 'Democratic theory: A preliminary mathematical model.'. *Public Choice*, 21, 99–103.
- Grofman, B. (1978). Judgmental competence of individuals and groups in a dichotomous choice situation: Is a majority of heads better than one? *Journal of Mathematical Sociology*, 6, 47–60.
- Grofman, B., & Feld, S. L. (1986). Determining optimal weights for expert judgement. In B. Grofman, & G. Owen (Eds.), *Information Pooling and Group Decision Making* (pp. 167–172). Bingley: Emerald.

- Grofman, B., & Owen, G. (1986). Condorcet models, avenues for future research. In B. Grofman, & G. Owen (Eds.), *Information Pooling and Group Decision Making* (pp. 93–102). Bingley: Emerald.
- Imai, K., King, G., & Lau, O. (2009). Zelig: Everyone's Statistical Software. URL: <http://gking.harvard.edu/zelig>.
- Kazmann, R. G. (1973). Democratic organization: A preliminary mathematical model. *Public Choice*, 16, 17–26.
- Ladha, K. K. (1992). The Condorcet jury theorem, free speech, and correlated votes. *American Journal of Political Science*, 36, 617–634.
- Lewis-Beck, M. S., & Skalaban, A. (1989). Citizen forecasting: Can voters see into the future? *British Journal of Political Science*, 19, 146–153.
- Lewis-Beck, M. S., & Stegmaier, M. (2011). Citizen forecasting: Can UK voters see the future? *Electoral Studies*, 30, 264–268.
- Lewis-Beck, M. S., & Tien, C. (1999). Voters as forecasters: A micromodel of election prediction. *International Journal of Forecasting*, 15, 175–184.
- Linzer, D. A. (2013). Dynamic Bayesian forecasting of presidential elections in the states. *Journal of the American Statistical Association*, 108, 124–134.
- Miller, M. K., Wang, G., Kulkarni, S. R., Poor, H. V., & Osherson, D. N. (2012). Citizen forecasts of the 2008 US presidential election. *Politics & Policy*, 40, 1019–1052.
- Mirkin, B. (1979). *Group Choice*. Washington, D.C: V. H. Winston.
- Murr, A. E. (2011). 'Wisdom of crowds'? A decentralised election forecasting model that uses citizens' local expectations. *Electoral Studies*, 30, 771–783.
- Murr, A. E. (2013). *Citizen Forecasting in the 2010 British General Election*. PhD Thesis, University of Essex.
- Nitzan, S., & Paroush, J. (1982). Optimal decision rules in uncertain dichotomous choice situations. *International Economic Review*, 23, 289–97.
- Rothschild, D., & Wolfers, J. (2013). Forecasting elections: Voter intentions versus expectations. *Working Paper*, Last Accessed: 9 March 2014. URL: <http://users.nber.org/~jwolfers/Papers/VoterExpectations.pdf>.
- Shapley, L., & Grofman, B. (1984). Optimizing group judgmental accuracy in the presence of interdependencies. *Public Choice*, 43, 329–343.

- Thompson, S. K. (1992). *Sampling*. New York & Chichester: Wiley.
- Thompson, S. K., & Seber, G. A. F. (1996). *Adaptive sampling*. New York & Chichester: Wiley.
- Ward, M. D., Greenhill, B. D., & Bakke, K. M. (2010a). The perils of policy by p-value: Predicting civil conflicts. *Journal of Peace Research*, 47, 1–13.
- Ward, M. D., Greenhill, B. D., & Bakke, K. M. (2010b). The perils of policy by p-value: Protecting civil conflicts. *Replication Data*, Last accessed: 10 March 2014. URL: [http://file.prio.no/Journals/JPR/2010/47/4/Ward%20GB%20Replication%20Files%2047\(4\).zip](http://file.prio.no/Journals/JPR/2010/47/4/Ward%20GB%20Replication%20Files%2047(4).zip).